

04

Control regions

In red: not applied

In green: applied in phase 0 or 1

Control regions: $t\bar{t}h$

	Invariant mass (GeV)	acoplanarity	η	2 isolated ts	N of b-jets	p_t
DY	[40,100]	<0.3	<2.4	Opposite sign	0	
QCD	[100,300]	>0.8	<2.4	Same sign	0	<75
ttjets	[200,650]	>0.5	<2.4	Opposite sign	1 or more	<125

samples used and information

Background	Dataset	Cross section (pb)
Drell Yan (DY)	/DYJetsToLL_M*_TuneCP5_13TeV-amcatnloFXFX-pythia8/RunIISummer20UL18NanoAODv9-106X*/NANOAODSIM	6404
inclusive $t\bar{t}$	TTJets_TuneCP5_13TeV-amcatnloFXFX-pythia8/NANOAODSIM/106X_upgrade2018_realistic_v16_L1v1-v1	832
QCD, $e\tau_h$ channel	Run2018*/EGamma/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v1	data-driven – luminosity 54.9 fb^{-1}
QCD, $\mu\tau_h$ channel	Run2018*/SingleMuon/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v2	data-driven – luminosity 54.9 fb^{-1}

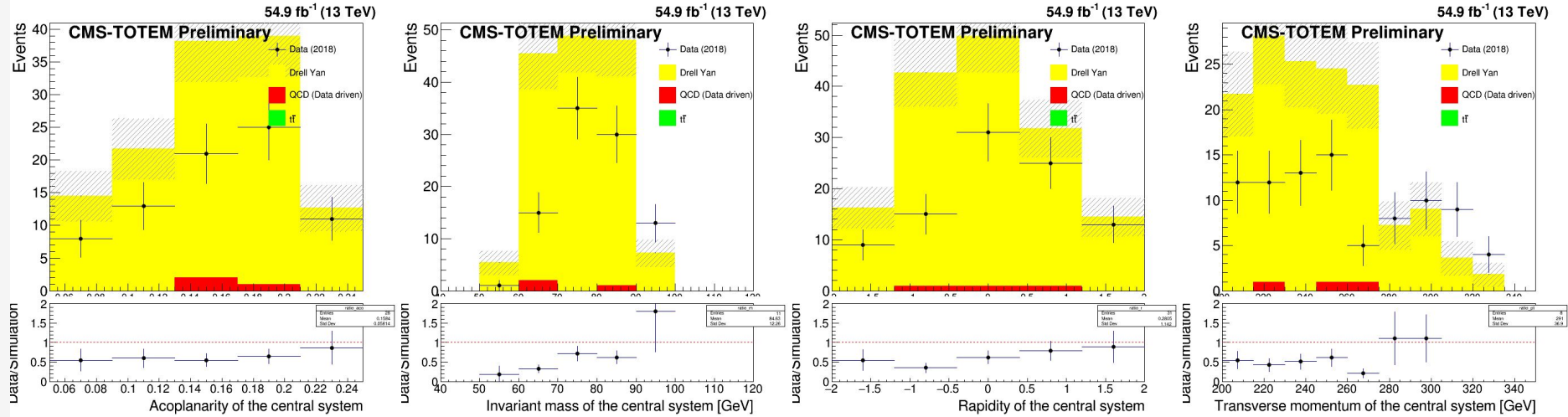
Table 3: List of the background considered. As for the Drell-Yan (DY), only events with M_X in the range $[50, 3000]$ GeV are considered (according to PPS acceptance [6]). For QCD, a data-driven approach is considered and data are used. This table refers to the 2018 data-taking period.

Questions

- The samples I am using (the ones mentioned in the twiki) come from the same datasets (referenced in the previous slide)?
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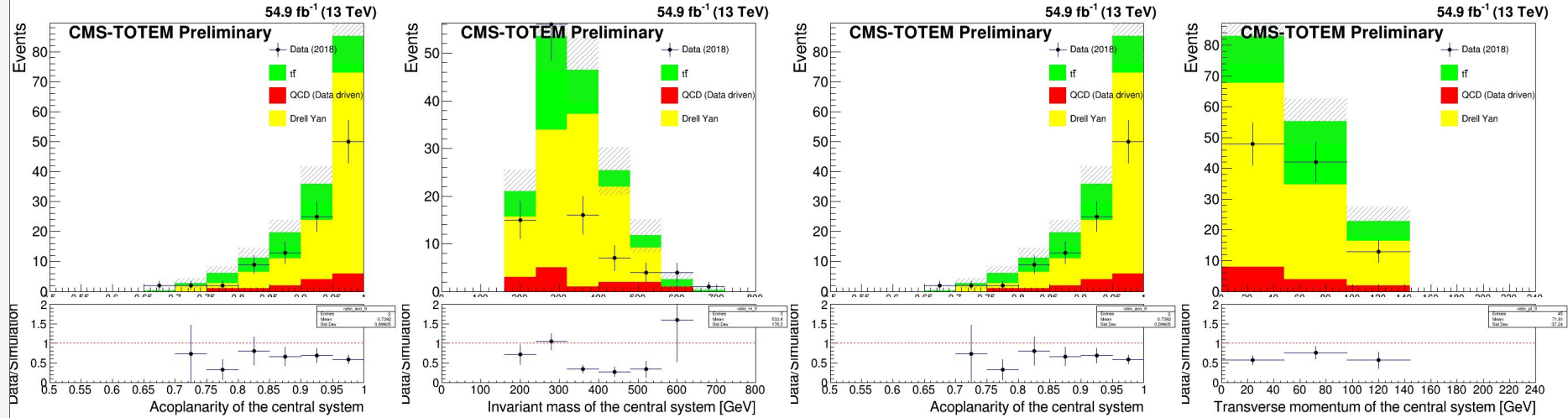
Control region plots - DY control region

tau mass [40,100]
 system acoplanarity <0.3
 n° of b jets =0



TT Control region

tau mass [200,650]
 system acoplanarity > 0.5
 n° of b jets > 0
 tau pair pt < 125



QCD control region

tau mass [100,400]
 system acoplanarity > 0.8
 n° of b jets = 0
 tau pair pt < 75

